SMART INDIA HACKATHON 2026

CIPHER VISION

- **Problem Statement ID** – 26171
- **Problem Statement** – On-device Visual Perception for Light-weight Browser Agents
- **Organization** – Indian Space Research Organisation (ISRO)
- **Theme** – Smart Automation
- **PS Category** – Software
- **Team ID** – [Enter Team ID]
- **Team Name** – CipherHex



WHY "CIPHER VISION"?

- ❑ *Cipher* → redaction/privacy-by-architecture core
- ❑ *Vision* → on-device visual perception engine reading the screen like a human would, safely

CipherHex

CIPHER VISION

A privacy-first browser agent that sees and acts on your screen entirely on-device-detecting and redacting sensitive data in real time, so no screen content ever leaves your devices unless it truly needs to.

SMART INDIA HACKATHON 2026

PROPOSED SOLUTION:

1 Phase 1

Goal decomposition

- Scrapes page into context
- On-device model breaks goal into ordered checklist

2 Phase 2

Perception

- Reads DOM + PII state
- Decision engine picks fastest sufficient tier

3 Phase 3

Execution

- Resolved actions run via event simulation
- Reliable even on dynamic, React-based UIs

4 Phase 4

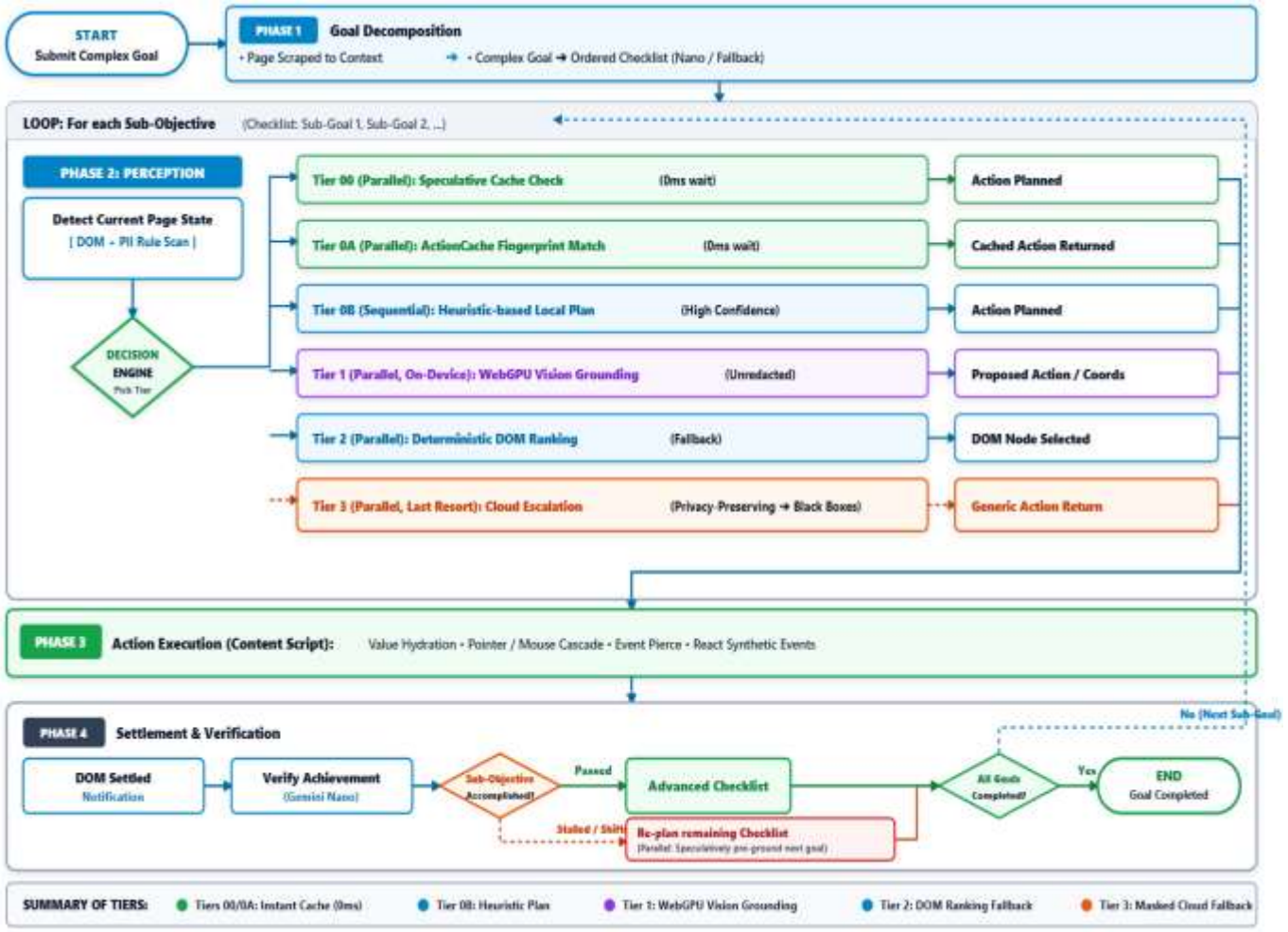
Settlement & verification

- Waits for DOM to settle, verifies sub-objective
- Auto re-plans checklist until goal is complete

KEY INNOVATION:

Unlike fixed pipelines, CipherVision dynamically adapts its speed-vs-privacy trade off per action – escalating to cloud only when local tiers can't resolve it, and only after redaction.

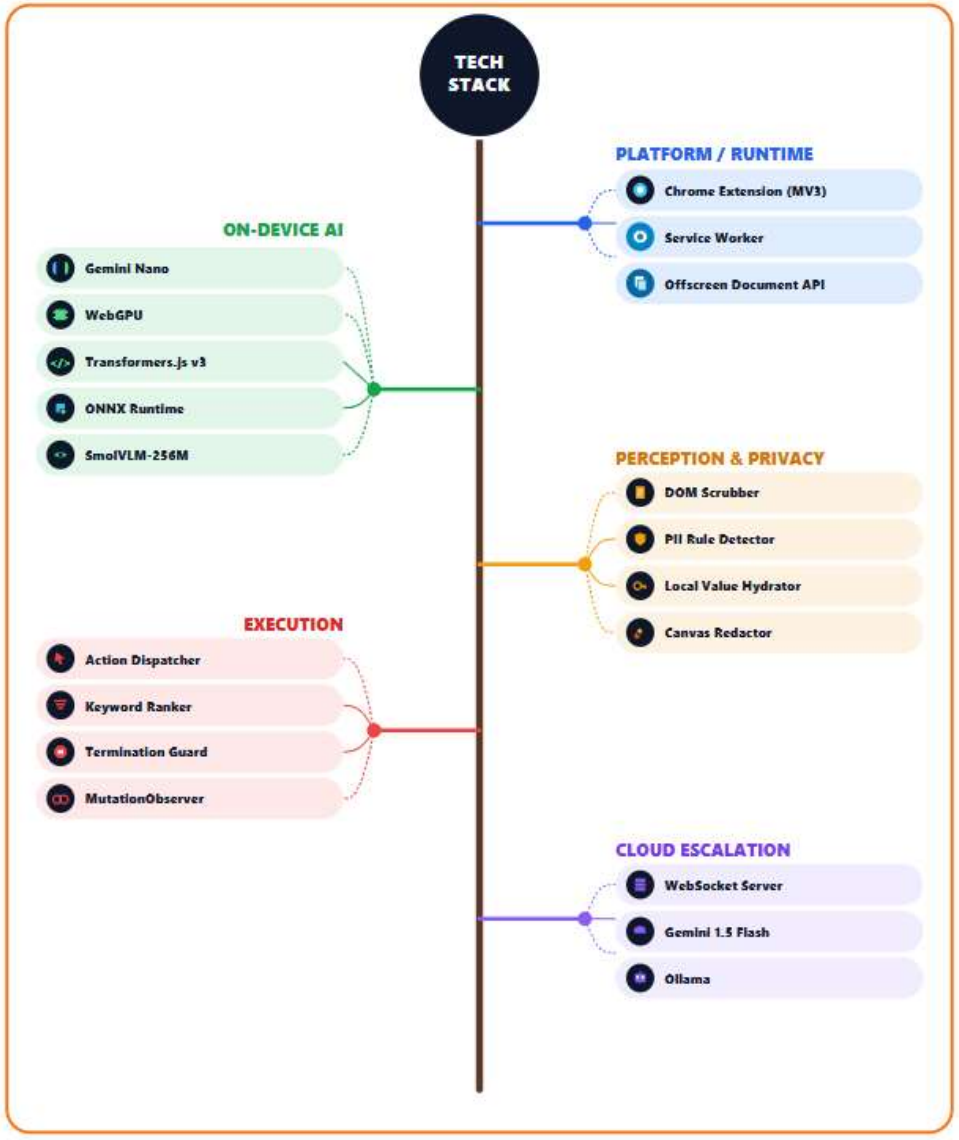
PROJECT FLOW



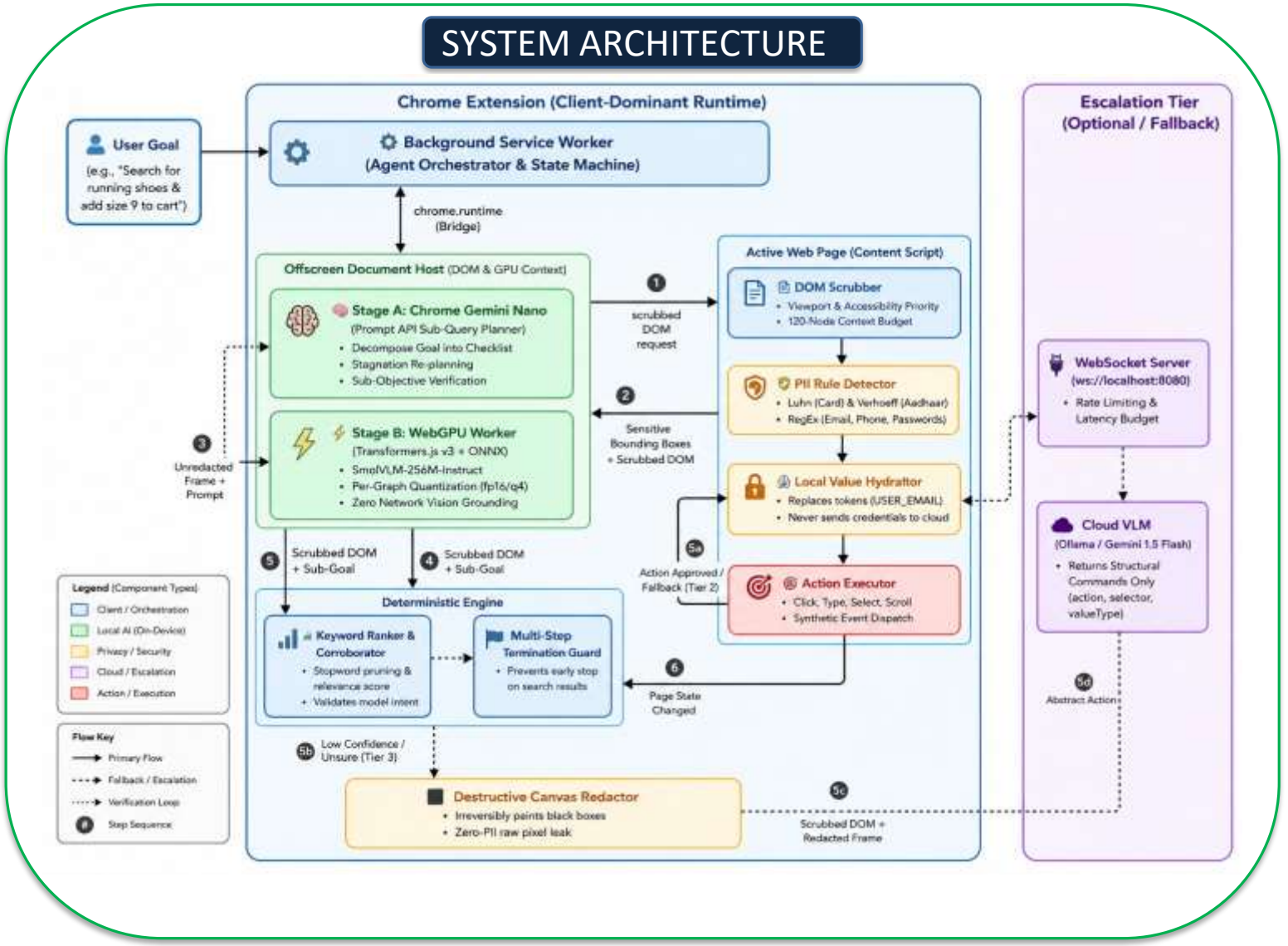
CipherHex

TECHNICAL APPROACH

SMART INDIA HACKATHON 2026



SYSTEM ARCHITECTURE



CipherHex

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

WHY IT'S FEASIBLE?

- **Technical Feasibility:** Built on browser APIs that already ship today, no experimental or unreleased tech required.
- **Cost Feasibility:** High-confidence actions resolve entirely on-device, with only redacted JSON (never raw screens) sent on escalation.
- **Operational Feasibility:** System degrades safely by design – low confidence escalates, ambiguous data over redacts, rather than failing or leaking.
- **Communication Feasibility:** Server contract is fixed and minimal – cloud only ever receives structured, sanitized context and only ever returns action instructions, never free-form data.

SCOPE

- **Prevent Data Leakage:** Guarantee no raw PII or screen content ever leaves the device unredacted — the core privacy contract the entire system is built around.
- **Minimize Server Dependency:** Resolve high-confidence actions entirely on-device, escalating to the cloud only when local reasoning genuinely cannot decide.
- **Enable Safe Automation on Sensitive Workflows:** Let users automate tasks on forms, logins, and personal data without trading away privacy — the actual use case that justifies this architecture over a simpler cloud agent.

POTENTIAL CHALLENGES

→ **PII in non-standard formats:** Unlabelled fields or PII inside images/canvas can slip past rule-based detection.
→ **Canvas-rendered UIs have no DOM to read:** Dashboards, SPAs, and drawing tools often have no inspectable DOM structure
→**Local model confidence miscalibration:** A model that's overconfident on an unfamiliar page could act incorrectly without escalating; one that's underconfident escalates too often, defeating the privacy by default design.
→ **Escalation adds latency**: Cloud reasoning introduces network round-trip time on complex tasks.

OVERCOMING CHALLENGES

→ Layer OCR/visual fallback; default to redaction when unsure.

→ Fall back to local vision-based screen reading for these cases.

→ Conservative default thresholds tuned toward escalation-on-doubt; termination guard halts and re-evaluates rather than repeating a wrong action.
→ Escalation is the exception, not default — most actions resolve locally

CipherHex

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

IMPACTS

PRIVACY

Raw screen/context → Blocked from cloud

Sensitive data is redacted at the source;
only sanitized state ever leaves the device.

ZERO RAW PIXEL EGRESS

LATENCY

Local resolution → Avoids network round-trip

Actions resolved on-device skip the round-trip
entirely, cutting response time.

SUB-50ms RESPONSE

CLOUD DEPENDENCY

Escalation trigger → Only when local tiers fail

Cloud is a fallback path, not the default —
invoked only when local tiers cannot resolve.

AIR-GAP OPERATIONAL

RELIABILITY

Closed-loop control → Verify → Fail → Replan

Every action is checked against the DOM;
failures trigger dynamic re-plan, not stalls.

SELF-HEALING LOOP

BENEFITS

- **Zero Raw Data Egress**
  Passwords, IDs, and session inputs never leave browser unredacted.
- **Sub-50ms Response on Simple Tasks**
  Local resolution eliminates network RTT for everyday navigation steps.
- **Transparent & Auditable Execution**
  Redaction stages are observable in real time via standard DevTools.
- **Air-Gap & Offline Functional**
  Tiers 0–2 execute locally without active connection or server dependency.
- **Zero-Cost Default Execution**
  Minimizing cloud escalation brings recurring API token expenses to ~$0.

KEY INNOVATION

Tier 0 (0ms): Instant Cache Hit & Structural Hash Replay

Tier 1 (<250ms): On-Device WebGPU Vision Grounding (SmolVLM-256M)

Tier 2 (<50ms): Spatial Accessibility Tree (AXTree) Heuristic Fallback

Tier 3 (Last Resort): Privacy-Preserving Cloud Escalation (Canvas Black Boxes)

WHY IT'S BETTER?

- **Near-Zero Latency:** Instant cached replay for repeated operator workflows.
- **Local-First Default:** 100% on-device perception; cloud only when ambiguous.
- **Self-Healing Loop:** Detects UI stalls & dynamically re-plans sub-objectives.
- **Hardware-Guaranteed PII Safety:** Masked prior to any network hop.

CipherHex

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

REFERENCES

PROTOTYPE DEMO VIDEO

✓ **WebArena: A Realistic Web Environment for Building Autonomous Agents** : https://arxiv.org/abs/2307.13854

✓ **Mind2Web: Towards a Generalist Agent for the Web** : https://arxiv.org/abs/2306.06070

✓ **Chrome Built-in AI — Prompt API (Gemini Nano)** : https://developer.chrome.com/docs/ai/built-in

✓ **WebGPU Specification (on-device vision grounding)** : https://www.w3.org/TR/webgpu/

✓ **Computer Use / Agentic Tool Architecture — Anthropic** : https://docs.anthropic.com/en/docs/agents-and-tools/tool-use/computer-use-tool

NOVELTY IN IMPLEMENTATION

❖ Tiered fallback decision engine (cache → heuristic → on-device vision → cloud) picks the fastest sufficient tier per action instead of a single fixed AI call

❖ Speculative caching + fingerprint matching enables 0ms action resolution for repeated UI patterns

❖ On-device WebGPU vision grounding keeps most actions private, with cloud escalation only as a privacy-preserving last resort

❖ Screenshots are redacted to black boxes before any cloud call, ensuring PII never leaves the device unprotected

❖ Self-healing execution loop auto-detects stalled/shifted sub-goals and re-plans the checklist without user intervention

DEMO SCREENSHOT

